FREEDOM INTERNATIONAL SCHOOL

WORKSHEET-MCQ-SOLUTIONS

PHYSICS

CLASS XII

RAY OPTICS AND OPTICAL INSTRUMENTS

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1. If an object is placed at 10 cm in front of a concave mirror of focal length 15 cm, the magnification of image is

   (a) -1.5 (b) 1.5 (c) -3 (d) 3

   Ans: (d) 3

   Given, u = -10cm, f = -15cm (for concave mirror)

   By mirror formula, $\frac{1}{u}+\frac{1}{v}=\frac{1}{f}$

   $\therefore \frac{1}{v}=\frac{1}{f}-\frac{1}{u}=\frac{1}{-15}-\frac{1}{-10}=\frac{-1}{15}+\frac{1}{10}$

   $=\frac{1}{30}$

   Image position, v = 30 cm

   Magnification, $m=-\frac{v}{u}=-\frac{30}{(-10)}=3$

2. To a fish under water, viewing obliquely a fisherman standing on the bank of a lake, the man looks

   (a) taller than what he actually is (b) shorter than what he actually is

   (c) the same height as he actually is (d) depends on the obliquity

   Ans: (a) Taller than what he actually is

3. The time taken by the light to cross a glass of thickness 4 mm and refractive index (μ= 3), will be

   (a) 4×10^{-11} s (b) 16×10^{-11} s (c) 2×10^{-11} s (d) 8×10^{-10} s

   Ans: (a) 4×10^{-11} s

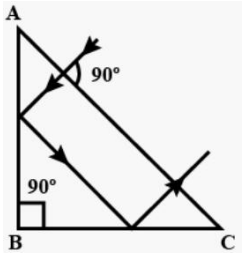
   μ= c/v

   v= c/μ= 3×10^8/ 3= 10^8 m/s

   Time= distance/ speed

   = 4×10^{-3}/ 10^8 = 4×10^{-11} s

4. A ray falls on a prism ABC (AB= AC) and travels as shown in the figure. The least value of refractive index material of the prism should be



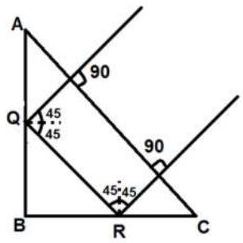
   (a) 1.5 (b) $\sqrt{2}$ (c) 1.33 (d) $\sqrt{3}$

Ans: (b) $\sqrt{2}$

$i_c = 45^0$

$n = 1/\sin i_c$

$n = \sqrt{2}$



5. Green light of wavelength 5460 A^0 is incident on an air- glass interface. If the refractive index of glass is 1.5, the wave length of light in glass would be (Given velocity of light in air= 3 x 10^8 m/s)

   (a) 3640 A^0 (b) 5460 A^0 (c) 4861 A^0 (d) 2640 A^0

   Ans: (a) 3640 A^0

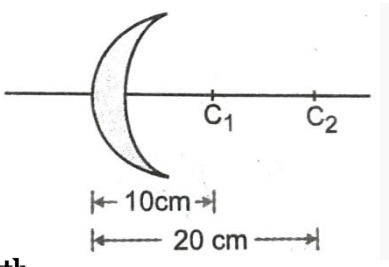
   $\frac{\lambda_1}{\lambda_2} = \frac{n_2}{n_1}$

   $\lambda_2 = \lambda_1 n_1/n_2 = 5460/\ 1.5 = 3640\ A^0$

6. The figure shows a thin lens with points C_1 and C_2 as the centres of curvature. The focal length of lens will be (refractive index=1.5)

   (a) 40 cm (b) 45 cm

   (c) 60 cm (d) 75 cm



   Ans:(a) 40 cm

   $R_1 = 10cm\ R_2 = 20cm\ \mu = 1.5$

   We can put these values in the lens makers formula to get the focal length,

   $\frac{1}{f} = (\mu - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$

   $\Rightarrow \frac{1}{f} = (1.5 - 1)\left(\frac{1}{10} - \frac{1}{20}\right)$

   $\Rightarrow \frac{1}{f} = \frac{1}{2} \times \frac{1}{20} = \frac{1}{40}$

   $\therefore f = 40cm$

7. How does the focal length of a convex lens change if monochromatic red light is used instead of violet light?

   (a) focal length is increased (b) focal length is decreased

   (c) focal length remains same (d) focal length does not depend on colour of light used

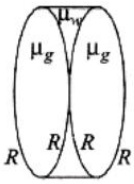
   Ans: (a) focal length is increased

   $f \alpha 1/(n-1)$. As $n_{red} < n_{violet}$ focal length will increase when red light is used.

8. Two identical glass ($\mu_g = 3/2$) equiconvex lenses of focal length f are kept in contact. The space between two lenses is filled with water ($\mu_w = 4/3$). The focal length of the combination is

   (a) f (b) f/2 (c) $\frac{4f}{3}$ (d) $\frac{3f}{4}$

   Ans: (d) $\frac{3f}{4}$



Here, $\mu_g = \frac{3}{2}$, $\mu_w = \frac{4}{3}$

Focal length (f) of glass convex lens is given by

$$\frac{1}{f} = (\mu_g - 1)\left(\frac{2}{R}\right)$$

or $\frac{1}{f} = \left(\frac{3}{2} - 1\right)\frac{2}{R} = \frac{1}{R}$ or $f = R$ ...(i)

Focal length (f) of water filled concave lens is given by

$$\frac{1}{f'} = (\mu_w - 1)\left(-\frac{2}{R}\right) \text{ or } \frac{1}{f'} = \left(\frac{4}{3} - 1\right)\left(-\frac{2}{R}\right)$$

$$= -\frac{2}{3R} = -\frac{2}{3f} \quad \text{[Using eqn. (i)]}$$

Equivalent focal length (f_{eq}) of lens system

$$\frac{1}{f_{eq}} = \frac{1}{f} - \frac{2}{3f} + \frac{1}{f} = \frac{3-2+3}{3f} = \frac{4}{3f}$$

$$\therefore \quad f_{eq} = \frac{3f}{4}$$

9. A ray incident at 15^0 on one refracting surface of a prism of angle 60^0, suffers a deviation of 55^0. What is the angle of emergence?

(a) 95^0 (b) 45^0 (c) 30^0 (d) 100^0

Ans: (d) 100^0

$$i + e = A + \delta$$

$$15^\circ + e = 60^\circ + 55^\circ$$

$$15^\circ + e = 115^\circ$$

$$e = 115^\circ - 15^\circ$$

$$e = 100^\circ$$

10. In order to increase the angular magnification of a simple microscope, one should increase

(a) the object size (b) the aperture of the lens

(c) the focal length of the lens (d) the power of the lens

Ans: (d) the power of the lens

11. For compound microscope f_o= 1cm, f_e= 2.5 cm. An object is placed at a distance 1.2 cm from objective lens. What should be length of microscope for normal adjustment?

(a) 8.5 cm (b) 8.3 cm (c) 6.5 cm (d) 6.3 cm

Ans: (a) 8.5 cm

L= $v_0+u_e= v_0+f_e$

$$\frac{1}{v_o} = \frac{1}{f_o} + \frac{1}{u_o}$$

Substituting the known values:

$$\frac{1}{v_o} = \frac{1}{1} + \frac{1}{-1.2}$$

Calculating:

$$\frac{1}{v_o} = 1 - \frac{5}{6} = \frac{1}{6}$$

Thus:

$$v_o = 6\,\text{cm}$$

L= 6+ 2.5 = 8.5 cm

12. The magnifying power of a telescope is 9. When it is adjusted for parallel rays, the distance between the objective and the eyepiece is found to be 20 cm. The focal lengths of lenses are

(a) 18 cm, 2 cm (b) 11 cm, 9 cm (c) 10 cm, 10 cm (d) 15 cm, 5cm

Ans: (a) 18 cm, 2 cm

Here, $M = \frac{f_0}{f_e} = 9 \therefore f_0 = 9f_e$ .....(i)

$f_0 + f_e = 20cm$

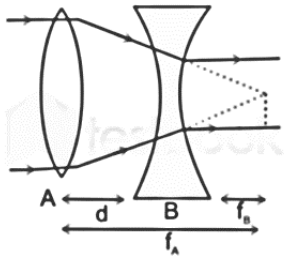
Using (i), $9f_e + f_e = 20$ or $f_e = 2cm$

From (i), $f_0 = 9 \times 2cm = 18cm.$

13. A convex lens A of focal length 20 cm and a concave lens of focal length 5 cm are kept along the same axis with a distance d between them. If a parallel beam of light falling on A leaves B as parallel beam, then the distance d (in cm) will be

(a) 25 (b) 15 (c) 30 (d) 50

Ans: (b) 15



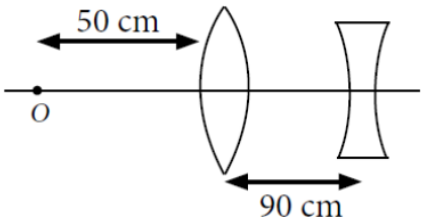
According to the above ray diagram, we can see that distance d is equal to,

$d = f_A - f_B$

$\Rightarrow d = 20 - 5$

$\Rightarrow d = 15$ cm

14. The focal length of a convex lens is 100 cm and focal length of concave lens is 8 cm. Find the magnification of the given combination of lens if object is placed at point O



(a) 1.02 (b) 0.04 (c) 2.01 (d) 0.08

Ans: (d) 0.08

For first lens

$$m_1 = \frac{v}{u} = \frac{f}{u+f} = \frac{100}{-50+100} = 2$$

Image distance v = 2u = 2 × (-50) = -100 cm

For second lens

Now u = - 190 cm, f = - 8 cm, m_2 = ?

$$m_2 = \frac{v}{u} = \frac{f}{u+f} = \frac{-8}{-190-8} = 0.04$$

Net magnification = $m_1 m_2$ = 0.08

15. An astronomical refracting telescope will have large angular magnification and high angular resolution, when it has an objective lens of

(a) small focal length and large diameter (b) small focal length and small diameter

(c) large focal length and large diameter (d) large focal length and small diameter

Ans: large focal length and large diameter

For questions 11 to 15, two statements are given- one labelled Assertion (A) and other labelled Reason (R). Select the correct answer to these questions from the options as given below.

A. Both Assertion and Reason are true and Reason is the correct explanation of Assertion.
B. Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.
C. Assertion is true but Reason is false.
D. Both Assertion and Reason are false.

16. **Assertion:** A diamond of refractive index $\sqrt{6}$ is immersed in a liquid of refractive index $\sqrt{3}$. If light travels from diamond to liquid, total internal reflection will take place when angle of incidence is 30^0.
Reason: $\mu = \sin i_c$, where μ is the refractive index of diamond with respect to the liquid.
Ans: D
17. **Assertion:** Refractive index of glass is more than the refractive index of air.
Reason: A double convex air bubble is formed within a glass slab. The air bubble behaves like a converging lens.
Ans: C
18. **Assertion:** If the objective lens and the eyepiece lens of a microscope are interchanged, it works as a telescope.
Reason: Objective lens of telescope requires small focal length and eyepiece lens require large focal length.
Ans: D
19. **Assertion:** The images formed by total internal reflections are much brighter than those formed by mirrors or lenses.
Reason: There is no loss of intensity in total internal reflection.
Ans: A
20. **Assertion:** The focal length of an equiconvex lens placed in air is equal to radius of curvature of either face.
Reason: For an equiconvex lens radius of curvature of both the faces is same.
Ans: B